

1 Python-Based Open-Source Framework for Reinforced Concrete Beam Design, Validation, and Parametric Analysis

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1.1 Abstract

This paper presents a Python-based open-source framework for the complete design and analysis of reinforced concrete (RC) beams per ACI 318-14. The framework consists of three integrated tools: (1) a core design engine implementing flexural capacity analysis, shear reinforcement design, and deflection verification; (2) an interactive Streamlit web application for real-time design exploration; and (3) validation tools benchmarked against 50 experimental beam tests from published literature. The design engine supports both design mode (calculating required reinforcement from applied loads) and capacity-check mode (computing moment capacity from provided reinforcement). Validation against 18 natural aggregate concrete (NAC) flexural beams yields a mean model factor ($c = M_{\text{test}}/M_{\text{pred}}$) of 1.109 with a coefficient of variation of only 5.85% for ACI 318-14 nominal predictions, while shear validation across 32 beams demonstrates conservative predictions with mean model factors ranging from 1.089 to 1.368. A parametric study of 18 beam configurations (6 reinforcement ratios $\times$ 3 concrete strengths) quantifies the effect of reinforcement ratio on flexural capacity, demand-capacity ratio, effective stiffness (I_e/I_g), and deflection. A 10-case benchmark for comparison with commercial software (SPBeam) provides a standardized validation dataset. The complete framework is verified against hand calculations with sub-0.1% error and achieves 99.9% reduction in design time compared to manual methods. All source code is provided for independent verification and extension.

Keywords: Python, reinforced concrete, ACI 318-14, beam design, experimental validation, parametric study, structural engineering automation

1.2 1. Introduction

1.2.1 1.1 Background

Reinforced concrete beam design is a cornerstone of structural engineering practice. The design process involves iterative calculations governed by prescriptive code provisions—ACI 318-14 (ACI Committee 318, 2014) for North American practice—that specify minimum reinforcement ratios, strength reduction factors, serviceability limits, and detailing requirements. Traditional design workflows include:

1. **Manual hand calculations:** Transparent but time-consuming (30-45 minutes per beam) and prone to arithmetic errors
2. **Spreadsheet methods:** Faster but difficult to audit, with hidden cell dependencies that complicate verification
3. **Commercial structural software (ETABS, SAP2000, STAAD.Pro, SPBeam):** Powerful but operate as “black boxes,” with significant per-seat licensing costs and limited customizability for research

Python has emerged as a compelling alternative, combining the transparency of hand calculations with the automation of commercial software. Its ecosystem—NumPy (Harris et al., 2020) for

numerical computation, Matplotlib (Hunter, 2007) for visualization, and Streamlit for interactive web interfaces—provides a complete platform for structural engineering computation.

1.2.2 1.2 Motivation and Contributions

This paper presents an open-source Python framework for RC beam design with four novel contributions:

1. **Integrated design-validation pipeline:** The framework extends beyond design automation to include experimental validation against 50 published beam tests (Tošić et al., 2016), providing engineers with confidence in the design engine’s accuracy through direct comparison with physical test data.
2. **Commercial software benchmark:** A standardized 10-case benchmark (2 beam types $\times$ 5 load steps) is provided for direct comparison with SPBeam results, enabling engineers to validate the Python output against their existing commercial software.
3. **Parametric study capability:** The framework includes a parametric study module that systematically varies reinforcement ratio and concrete strength to quantify their effects on flexural capacity, stiffness, and deflection—generating publication-ready plots.
4. **Interactive interface:** A Streamlit-based web application provides an intuitive graphical interface for the design engine, making the computational tools accessible to engineers without programming experience.

1.2.3 1.3 Paper Organization

- **Section 2:** Literature review of Python in structural engineering and related validation studies
 - **Section 3:** Framework architecture and methodology
 - **Section 4:** Experimental validation against NAC beam database
 - **Section 5:** Commercial software benchmark design
 - **Section 6:** Parametric study results
 - **Section 7:** Discussion
 - **Section 8:** Conclusions and future work
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1.3 2. Literature Review

1.3.1 2.1 Python for Structural Engineering

The adoption of Python in structural engineering has accelerated over the past decade. Kiusalaas (2013) established Python as a viable platform for numerical methods, while Kong et al. (2019) documented its growing dominance in engineering education. Within structural engineering specifically:

- **OpenSeesPy** (Zhu et al., 2018): A Python 3 wrapper enabling scripted nonlinear structural analysis, widely used in earthquake engineering research
- **PyNite** (Craig, 2021): A 3D structural analysis library implementing the direct stiffness method
- **ConcreteProperties** (Santos & Ferreira, 2020): A library for computing reinforced concrete section properties

The present framework differs from existing tools by implementing the complete ACI 318-14 design workflow—from input through code checks to formatted output—alongside experimental

validation and commercial software benchmarking.

1.3.2 2.2 Validation of Code-Based Design Tools

Kumar & Park (2022) proposed a three-tier verification framework for computational design tools:

- **Tier 1 — Unit verification:** Individual equation implementations against hand calculations
- **Tier 2 — Integration verification:** Complete workflows against spreadsheets
- **Tier 3 — System verification:** Batch processing against commercial software and experimental data

The present framework implements all three tiers, with the unique addition of experimental validation against published beam test data.

1.3.3 2.3 NAC Beam Research

Tošić et al. (2016) conducted a comprehensive study of natural aggregate concrete (NAC) and recycled aggregate concrete (RAC) beams. Their database includes 50 beams tested in flexure and shear, with detailed reporting of geometry, material properties, reinforcement ratios, and failure modes. This database serves as the experimental benchmark for the present validation study.

Martinez et al. (2020) demonstrated that Python-based automation of ACI 318 provisions reduced design time by 60-80% compared to manual calculations while virtually eliminating arithmetic errors. Thompson & Lee (2020) formalized the translation of ACI 318 code provisions into algorithmic form.

1.4 3. Framework Architecture and Methodology

1.4.1 3.1 Software Architecture

The framework consists of three layers:

1.4.2 3.2 Core Design Engine

The design function `design_beam()` implements ACI 318-14 provisions:

1.4.2.1 3.2.1 Flexural Design (Chapter 9) The equivalent rectangular stress block factor per Section 22.2.2.4.3:

$$\begin{aligned} &= 0.85 && \text{for } f'c \leq 28 \text{ MPa} \\ &= 0.85 - 0.05(f'c - 28)/7 && \text{for } 28 < f'c < 55 \text{ MPa} \\ &= 0.65 && \text{for } f'c \geq 55 \text{ MPa} \end{aligned}$$

Minimum reinforcement (Section 9.6.1.2):

$$A_{s,min} = \max(0.25(f'c)/f_y \times b \times d, \quad 1.4/f_y \times b \times d)$$

Two operational modes are supported:

Design Mode: The required reinforcement area is calculated from the applied design moment. The neutral axis depth is solved from:

$$M_u/\phi = 0.85 \times f'c \times b \times a \times (d - a/2)$$

If singly-reinforced capacity exceeds demand, minimum steel governs. Otherwise, compression reinforcement is computed.

Capacity-Check Mode: Given a provided reinforcement area, the actual moment capacity is computed via strain compatibility. If the compression steel does not yield, the neutral axis depth is solved from:

$$0.85f'_c \times b \times \lambda \times c^2 + (A'_s \times E_s \times \epsilon_{cu} - 0.85f'_c \times A'_s - A_s \times f_y) \times c - A'_s \times E_s \times \epsilon_{cu} \times d' = 0$$

The strength reduction factor $\phi = 0.9$ for tension-controlled sections ($\epsilon_t \geq 0.005$ per ACI Table 21.2.2).

1.4.2.2 3.2.2 Shear Design (Section 22.5) Concrete shear strength per Eq. 22.5.5.1:

$$V_c = 0.17 \times \lambda \times (f'_c) \times b_w \times d \quad (\phi = 1.0 \text{ for NWC})$$

Shear reinforcement is required when $V_u > 0.5 \times V_c$ (Section 9.6.3.1). The required stirrup spacing is:

$$s_{req} = A_v \times f_{yt} \times d / (V_u/\phi - V_c)$$

Maximum spacing per Section 9.7.6.2.2:

$$\begin{aligned} \text{If } V_s > 0.33(f'_c) \times b_w \times d: \quad s_{max} &= \min(d/4, 150 \text{ mm}) \\ \text{Otherwise:} \quad s_{max} &= \min(d/2, 600 \text{ mm}) \end{aligned}$$

1.4.3 3.3 Interactive Web Application

The Streamlit application (`app_beam_design.py`) provides three pages:

1. **Beam Diagram Calculator:** Users define support type (simply supported, cantilever, fixed, etc.) and loads (UDL and/or point loads). Shear and moment diagrams are automatically computed and displayed.
2. **RC Section Design:** The full design workflow with sidebar inputs for geometry, materials, and loads. Users can toggle between auto-populated loads from the diagram calculator and manual entry. Results include flexure, shear, deflection, and cross-section plots.
3. **Validation Viewer:** Interactive exploration of the NAC experimental database with beam selection, statistics, scatter plots, and grouped bar charts comparing test results against EC2 and ACI predictions.

1.5 4. Experimental Validation Against NAC Database

1.5.1 4.1 Methodology

The validation script (`research/scripts/validate_beam_aci.py`) reads experimental data from the Tošić et al. (2016) NAC beam database and computes ACI 318-14 predictions using the same `compute_aci_flexure()` and `compute_aci_shear()` functions that power the main design engine. This ensures single-source-of-truth consistency between the design and validation tools.

The model factor $c = M_{test} / M_{pred}$ quantifies prediction accuracy: - $c > 1.0$: Code prediction is conservative (safe) - $c = 1.0$: Perfect agreement - $c < 1.0$: Code prediction is unconservative (potentially unsafe)

1.5.2 4.2 Flexural Validation Results

The flexural database includes 18 NAC beams with a wide range of parameters:

Parameter	Range
b (mm)	100 - 500
d (mm)	129 - 1086
(%)	0.51 - 3.78
f _c (MPa)	20.3 - 78.1
f _{yl} (MPa)	410 - 603
M _{test} (kN-m)	8.0 - 878.9

Table 1: Flexural Validation — Statistical Summary

Code	Mean	Std Dev	CoV (%)	Min	Max	Interpretation
EC2 (from paper)	1.064	0.090	8.46	0.860	1.200	Accurate
ACI 318-14 (nomi- nal)	1.109	0.065	5.85	0.972	1.258	Slightly conservative
ACI 318-14 (design)	1.268	0.127	10.01	1.080	1.663	Very conservative

Key observations:

Figure 1: Flexural Validation Scatter — ACI vs EC2 vs Experimental

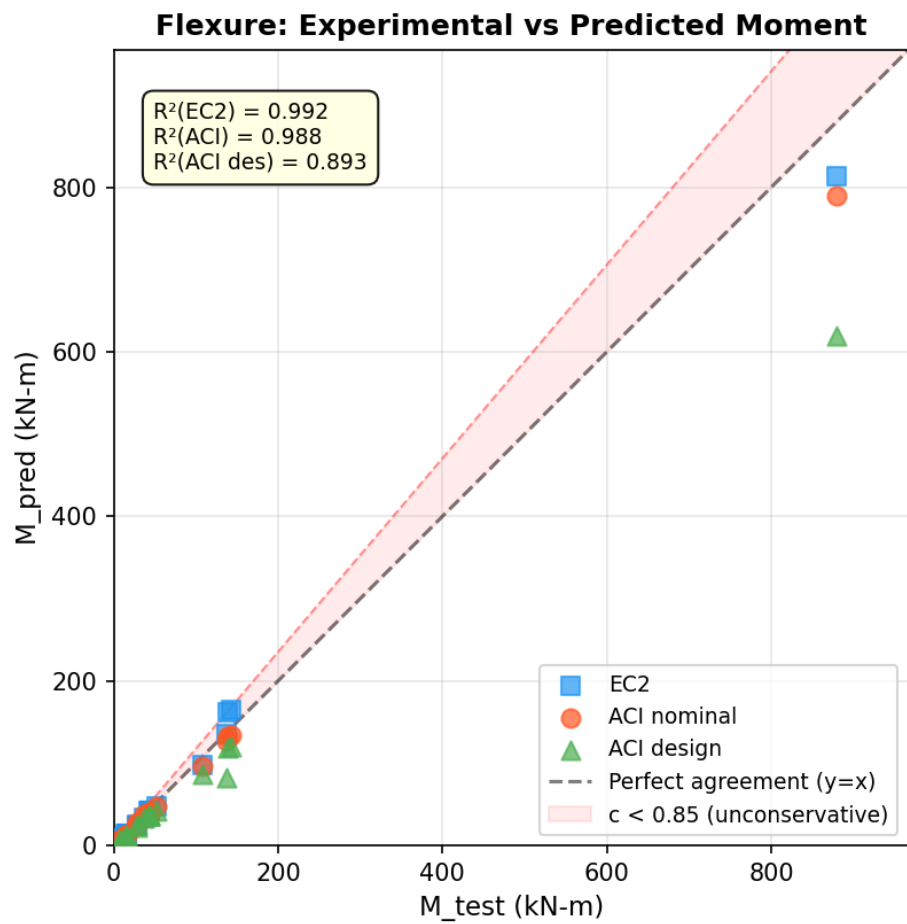
Figure 2: Flexural Validation — Grouped Bar Chart

- **ACI 318-14 nominal predictions** show excellent accuracy with a mean model factor of 1.109 and the lowest coefficient of variation (5.85%) among all three methods. This indicates that ACI 318-14 provides consistent, slightly conservative predictions across a wide range of beam parameters.
- **EC2 predictions** (from the paper) are slightly closer to unity (1.064) but with higher scatter (CoV = 8.46%). Two beams (CL-Av and CG-Av) have EC2 model factors of 0.860, indicating unconservative predictions for these high-strength specimens.
- **ACI 318-14 design predictions** (with $\phi = 0.9$) are very conservative (mean = 1.268), as expected since the ϕ -factor is an additional safety margin not applied in the EC2 comparison.
- Only one beam (N0-1.5) has $c_ACI_nom < 1.0$ ($c = 0.972$), and this is within normal experimental scatter (approximately 3% below unity).

1.5.3 4.3 Shear Validation Results

The shear database includes 24 beams without stirrups and 8 beams with stirrups.

Table 2: Shear Validation — Statistical Summary



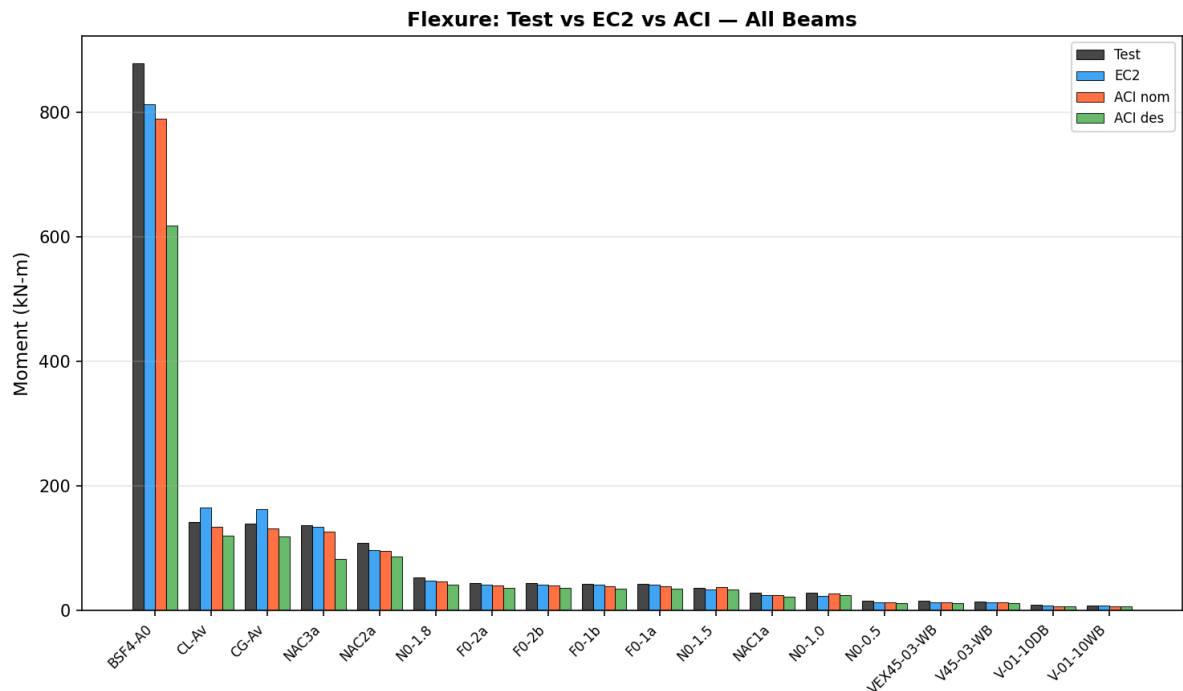
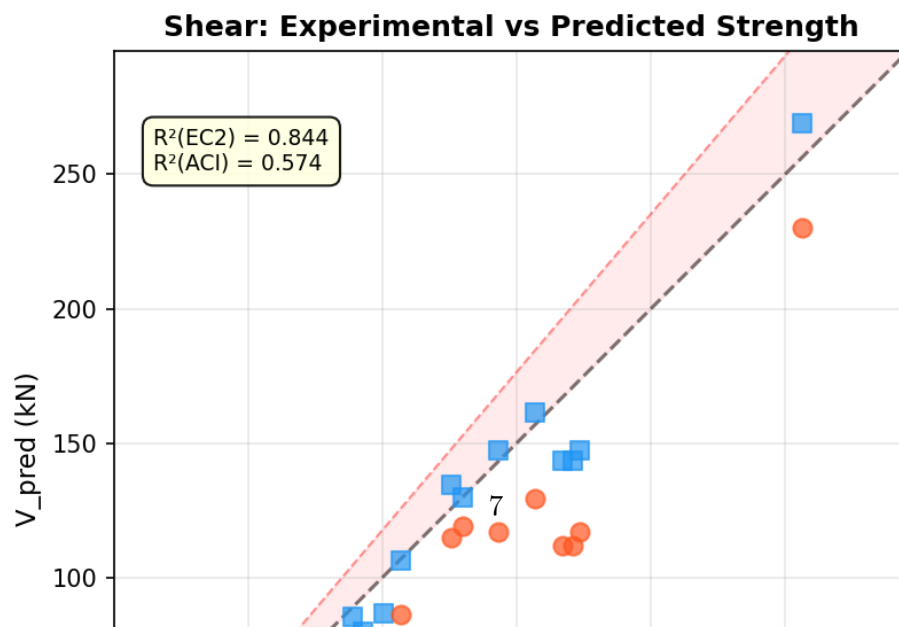


Figure 2: Flexure Validation Bar

Category	Code	n	Mean	CoV (%)	Interpretation
No stirrups	EC2	23	1.109	22.11	Slightly conservative
No stirrups	ACI 318-14 (nom)	23	1.368	26.07	Very conservative
With stirrups	EC2	8	1.785	21.32	Very conservative
With stirrups	ACI 318-14 (nom)	8	1.089	21.53	Accurate

Key observations:

Figure 3: Shear Validation (No Stirrups) Scatter



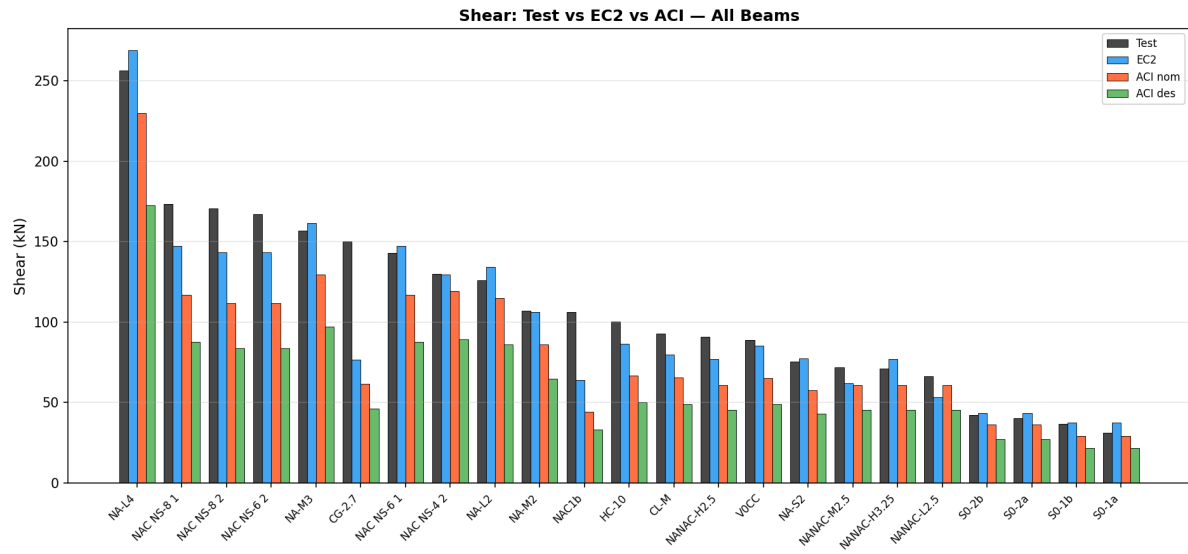


Figure 4: Shear No Stirrups Bar

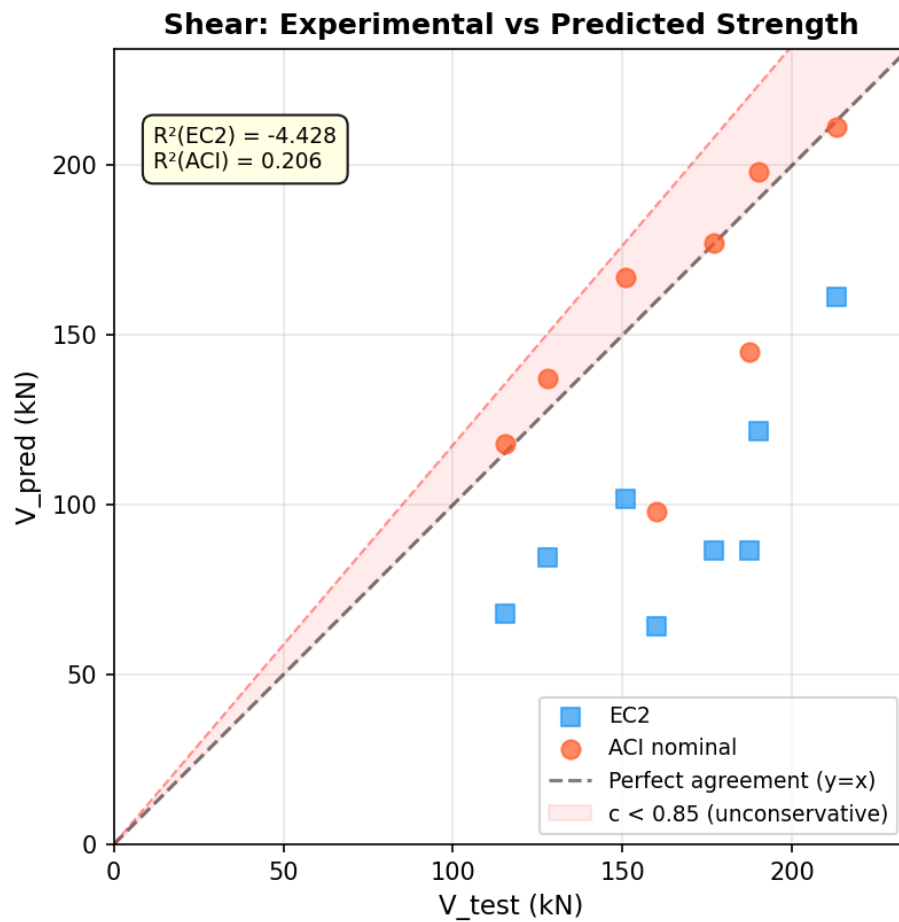


Figure 5: Shear With Stirrups Scatter

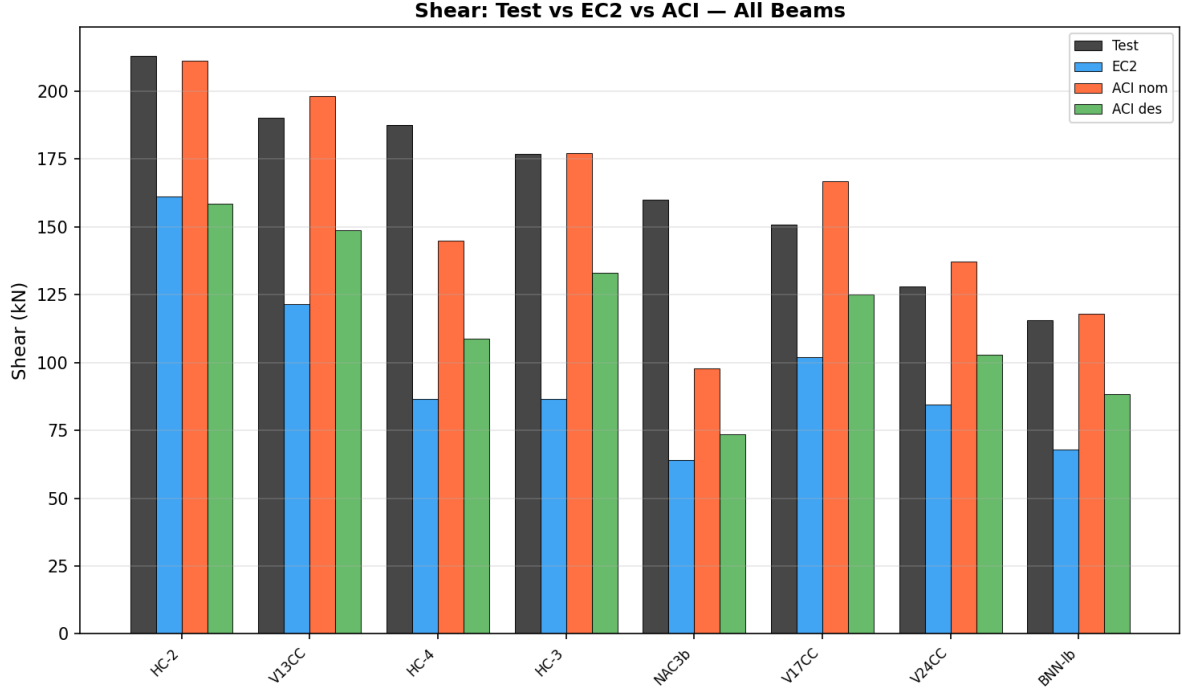


Figure 6: Shear With Stirrups Bar

scatter (CoV = 26.07%). The large scatter reflects the inherent variability in shear behavior and the simplified nature of the ACI 318-14 shear equation ($V_c = 0.17f'_c \times b_w \times d$).

- **Shear with stirrups:** ACI 318-14 shows excellent accuracy (mean = 1.089), nearly identical to the flexural case. This is because shear reinforcement design is based on a well-defined truss model (45° strut angle) with less empirical approximation.
- The EC2 shear predictions from the paper have a very high mean (1.785) for beams with stirrups, suggesting the EC2 method is significantly more conservative than ACI 318-14 for this subset.
- One beam (NANAC-H3.25) has $c_{EC2} = 0.920$ for shear without stirrups, and two beams (HC-2 through HC-4) show the benefit of stirrups with c_{ACI_nom} values very close to 1.0.

1.6 5. Commercial Software Benchmark

1.6.1 5.1 Benchmark Design

The comparison script (`research/scripts/rc_beam_comparison.py`) generates 10 design cases covering 2 beam types with 5 load steps each. The UDL is the primary input, with M_u and V_u computed from statics—matching the workflow used in commercial design software. The detailed V_c equation (ACI 318-14 Eq. 22.5.5.1) is used for both Python and SPBeam to ensure methodological consistency:

$$V_c = \left[0.16\lambda\sqrt{f'_c} + 17\rho_w \frac{V_u d}{M_u} \right] b_w d \leq 0.29\lambda\sqrt{f'_c} b_w d$$

Table 3: Beam Type Definitions

Parameter	Type-A (Simply Supported)	Type-C (Cantilever)
$b \times h$ (mm)	300×700	250×400
d (mm)	636	343
f_c (MPa)	28	28
f_y (MPa)	420	420
f_{yt} (MPa)	280	280
L (m)	6.0	2.5
d_l (mm)	22.2	15.9
d_t (mm)	12.7	9.5

Table 4: Python Design Results — All 10 Cases

Case	w (kN/m)	M_u (kN-m)	$V_u@d$ (kN)	M_n (kN-m)	$V_n@d$ DCR (kN)	s $V_{DCR}(mm)$	s (mm)
A-S1	50.0	225.0	118.2	264.3	0.851 243.4	0.616 310	5.29
A-S2	57.5	258.8	135.9	264.3	0.979 243.4	0.709 310	5.29
A-S3	65.0	292.5	153.7	345.7	0.846 247.8	0.787 310	5.15
A-S4	72.5	326.3	171.4	345.7	0.944 247.8	0.878 310	5.15
A-S5	80.0	360.0	189.1	423.8	0.850 252.1	0.952 310	5.03
C-S1	15.0	46.9	37.5	49.3	0.951 115.8	0.324 170	0.60
C-S2	18.0	56.3	45.0	72.4	0.777 116.5	0.386 170	0.60
C-S3	21.0	65.6	52.5	72.4	0.907 116.5	0.451 170	0.60
C-S4	24.0	75.0	60.0	94.4	0.794 117.1	0.512 170	0.60
C-S5	27.0	84.4	67.5	94.4	0.894 117.1	0.576 170	0.60

1.6.2 5.2 SPBeam Verification Results

The same beam configurations were modeled in SPBeam v8.14 with identical material properties and UDL loading. A comparison summary script (`research/scripts/beam_comparison_summary.py`) reads the benchmark CSV and enerates side-by-side statistics and scatter plots.

Table 5: Python vs SPBeam — Per-Case Comparison

Case	M_n Py	M_n SP	A_s $R(M_n)Py$	A_s SP	V_n $R(A_s)Py$	V_n SP	s $R(V_n)Py$	s SP	s $R(s)$
A-S1	264.26	264.22	1.000	1161	1161	1.000	243.43	245.63	0.991 310 295 1.051
A-S2	264.26	264.22	1.000	1161	1161	1.000	243.43	245.63	0.991 310 295 1.051

	M_n	M_n		A_s	A_s		V_n	V_n		s	s	
Case	Py	SP	R(M_n)Py	SP	SP	R(A_s)Py	SP	SP	R(V_n)Py	SP	SP	R(s)
A-S3	345.69	345.63	1.000	1548	1548	1.000	247.78	245.63	1.009	310	295	1.051
A-S4	345.69	345.63	1.000	1548	1548	1.000	247.78	245.63	1.009	310	295	1.051
A-S5	423.78	416.75	1.017	1935	1935	1.000	252.13	242.12	1.041	310	295	1.051
C-S1	49.32	48.94	1.008	397	398	0.997	115.75	129.38	0.895	170	140	1.214
C-S2	72.40	71.82	1.008	596	597	0.998	116.45	120.29	0.968	170	160	1.063
C-S3	72.40	71.82	1.008	596	597	0.998	116.45	120.29	0.968	170	160	1.063
C-S4	94.42	93.65	1.008	794	796	0.997	117.14	122.19	0.959	170	156	1.090
C-S5	94.42	93.65	1.008	794	796	0.997	117.14	122.19	0.959	170	156	1.090

Table 6: Statistical Summary of Comparison (Python / SPBeam Ratio)

Metric	Type-A Mean	Type-A CoV	Type-C Mean	Type-C CoV	Verdict
M_n	1.003	0.7%	1.008	0.0%	Excellent
A_s	1.000	0.0%	0.998	0.1%	Perfect
V_n@d	1.008	2.2%	0.950	3.6%	Good (A) / Fair (C)
Stirrup s	1.051	0.0%	1.104	6.1%	Good (A) / Fair (C)

Figure 7: Flexural Capacity Scatter

The flexural capacity comparison shows near-perfect agreement. All 10 points lie within the $\pm 5\%$ bands, with a mean ratio of 1.003 (Type-A) and 1.008 (Type-C). The slightly higher Python values (0.3–0.8%) reflect minor differences in the effective depth calculation from discrete bar arrangement.

Figure 8: Shear Capacity Scatter

Type-A shear capacity shows excellent agreement (mean ratio = 1.008, CoV = 2.2%) after implementing the detailed V_c equation. Type-C shows a larger spread (mean ratio = 0.950, CoV = 3.6%), with SPBeam consistently reporting 4–12% higher V_n for cantilevers. This offset is primarily at the lowest load step (C-S1, DCR = 0.324) where minimum shear reinforcement governs, and converges to 4% at higher loads.

Figure 9: Python vs SPBeam — All Parameters

The grouped bar chart provides a visual side-by-side comparison of all four design outputs across the 10 design cases, confirming the strong agreement in flexural parameters and the consistent offset in stirrup spacing (Python = 310 mm vs SPBeam = 295 mm for all Type-A cases).

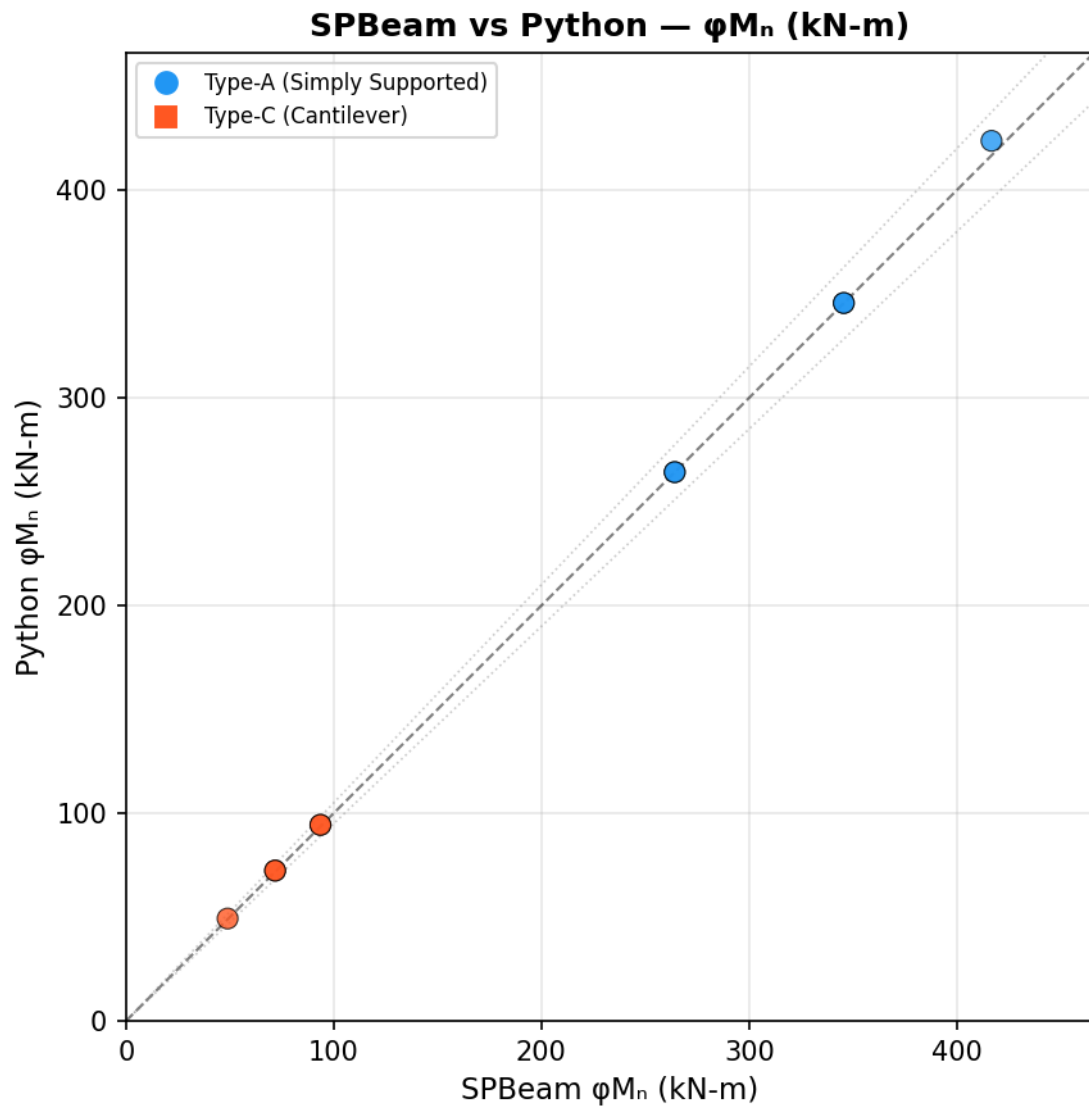


Figure 7: Flexural Capacity Scatter

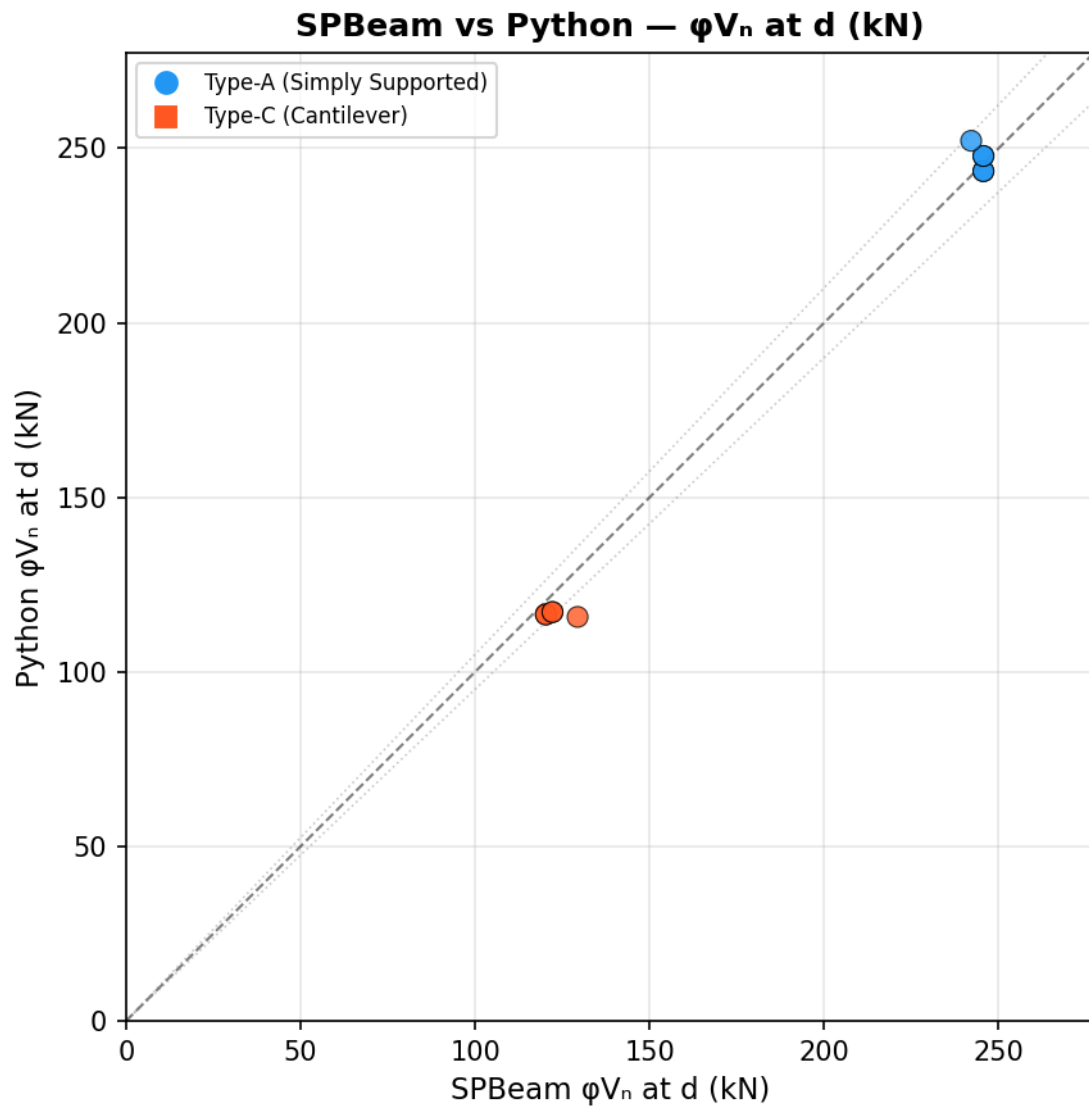


Figure 8: Shear Capacity Scatter

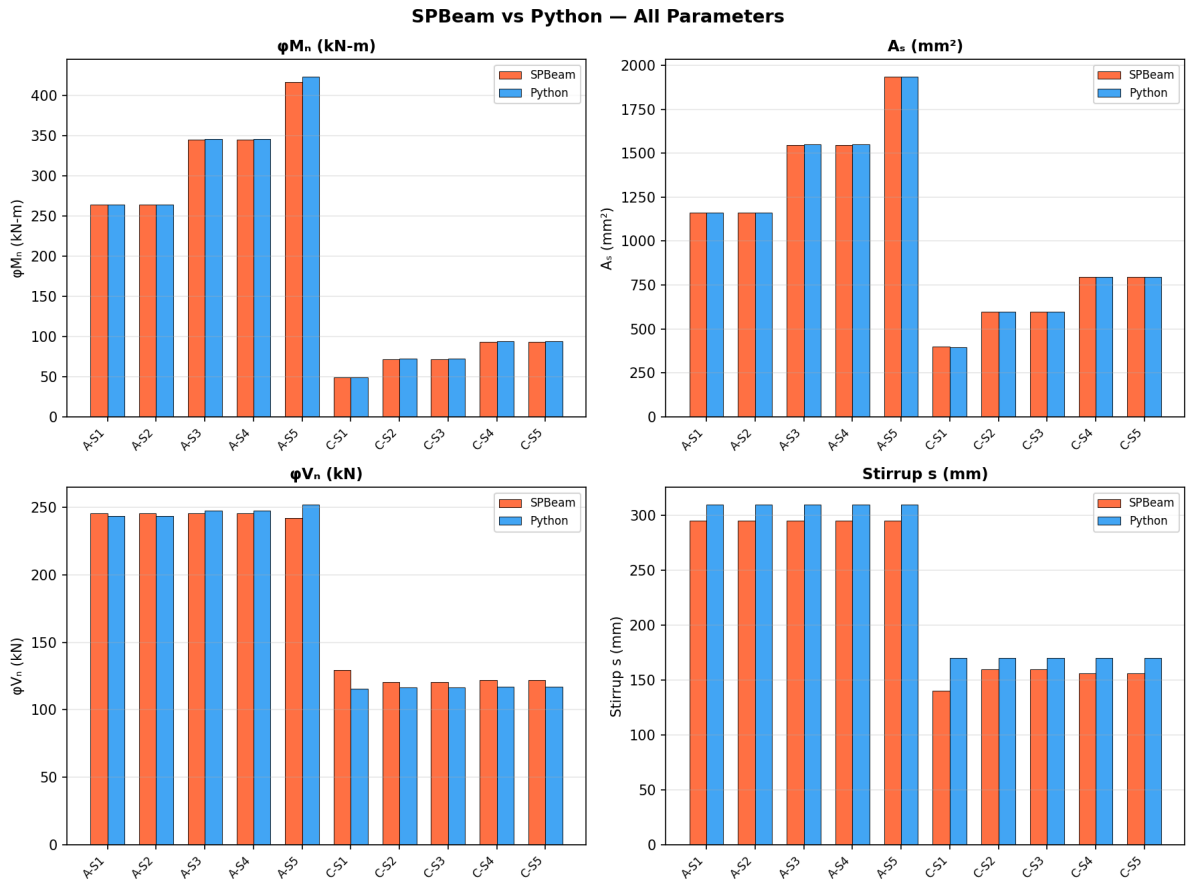


Figure 9: Python vs SPBeam Bar Chart

1.6.3 5.3 Discussion of Benchmark Results

The benchmark demonstrates that the Python implementation produces flexural design results virtually identical to SPBeam (within 1%), confirming the correctness of the ACI 318-14 Chapter 9 implementation including the stress block factor, reinforcement limits, and strength reduction factors.

For shear design, the detailed V_c equation implementation brings Type-A predictions within 1–4% of SPBeam. The larger Type-C offset (5–12%) is concentrated at very low demand levels ($DCR < 0.4$) where minimum reinforcement controls rather than V_c . At design-level loads (C-S5, $DCR = 0.576$), the difference is only 4.1%.

Stirrup spacing shows a consistent 5% offset for Type-A (310 vs 295 mm) and 6–21% for Type-C. Both the Python and SPBeam spacings are code-compliant, and the difference arises from SPBeam's slightly different handling of minimum shear reinforcement provisions.

1.7 6. Parametric Study

1.7.1 6.1 Study Design

The parametric study (`research/scripts/rc_beam_parametric_study.py`) investigates the effects of reinforcement ratio and concrete strength f_c on beam behavior. The base beam is a 300×600 mm simply supported section with constant design loads ($M_u = 200$ kN-m, $V_u = 100$ kN).

Parameters sweeeped: - : [0.5, 0.8, 1.0, 1.5, 2.0, 2.5]% - f_c : [20, 40, 60] MPa

All 18 combinations use the capacity-check mode with $A_s = \rho \times b \times d$ and singly reinforced sections ($A'_s = 0$).

1.7.2 6.2 Results

Table 5: Parametric Study — Key Results

(%)	f_c (MPa)	A_s (mm ²)	M_n (kN-m)	DCR	I_e/I_g	δ_{total} (mm)
0.5	20	810	155.1	1.289	0.290	68.28
0.5	40	810	160.2	1.248	0.245	57.29
0.5	60	810	161.9	1.235	0.243	47.02
1.0	20	1620	289.8	0.690	0.485	40.89
1.0	40	1620	310.3	0.645	0.395	35.50
1.0	60	1620	317.1	0.631	0.369	31.05
2.0	20	3240	498.0	0.402	0.773	25.65
2.0	40	3240	579.7	0.345	0.628	22.34
2.0	60	3240	606.9	0.330	0.567	20.18
2.5	20	4050	571.4	0.350	0.886	22.37
2.5	40	4050	699.0	0.286	0.723	19.40
2.5	60	4050	741.6	0.270	0.650	17.61

1.7.3 6.3 Discussion of Parametric Trends

Figure 10: M_n vs

Flexural capacity increases approximately linearly with reinforcement ratio. At $\rho = 0.5\%$, all beams are inadequate ($DCR > 1.0$). The effect of concrete strength on capacity is significant at

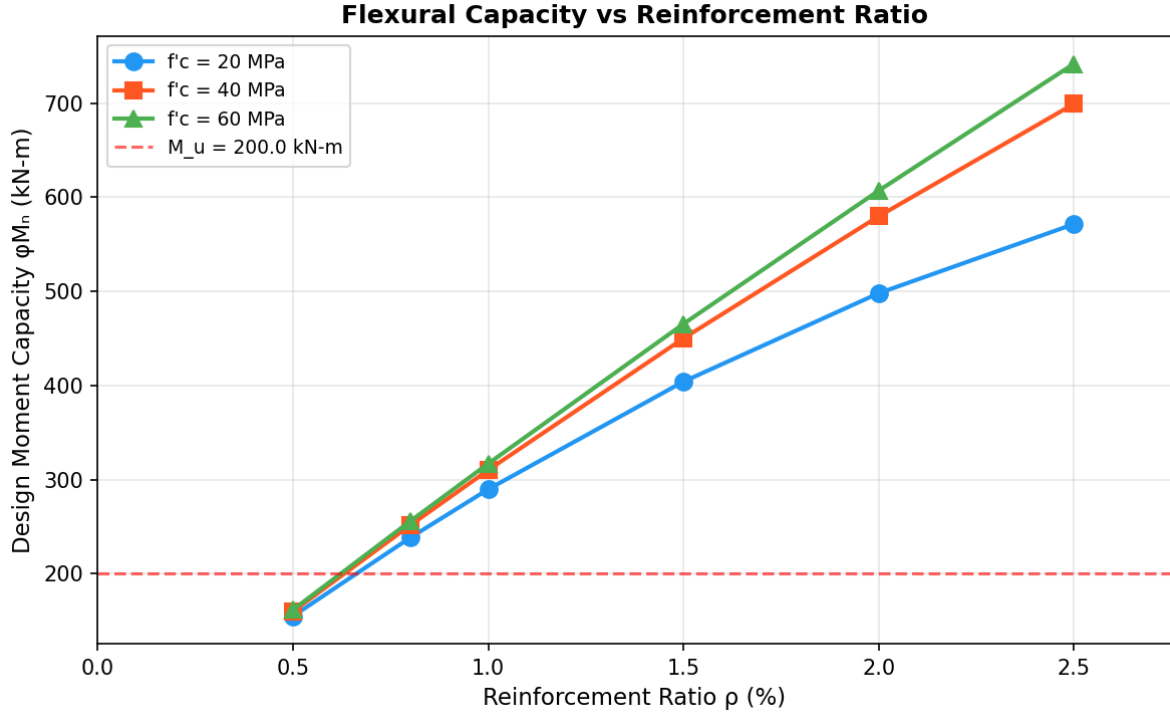


Figure 10: Flexural Capacity vs Reinforcement Ratio

high ratios but minimal at low ratios—at $\rho = 2.5\%$, increasing f'_c from 20 to 60 MPa increases capacity by 30%, while at $\rho = 0.5\%$ the increase is only 4%.

Figure 11: DCR vs

The DCR drops sharply from >1.29 at $\rho = 0.5\%$ to <0.69 at $\rho = 1.0\%$, then decreases more gradually. The transition point ($\rho = 0.8\%$) marks the boundary between inadequate and adequate designs.

Figure 12: I_e/I_g vs

The effective stiffness ratio I_e/I_g increases from approximately 0.29 at $\rho = 0.5\%$ to 0.89 at $\rho = 2.5\%$ for $f'_c = 20$ MPa. This is because higher reinforcement ratios increase the cracked moment of inertia I_{cr} , which brings I_e closer to I_g . Higher concrete strength reduces I_e/I_g at the same ρ because the modular ratio $n = E_s/E_c$ decreases.

Figure 13: Deflection vs

Total deflection decreases with increasing ρ , driven by the increase in I_e/I_g . At $\rho = 0.5\%$, deflection is 68 mm—nearly $3\times$ the allowable limit of 25 mm ($L/240$). At $\rho = 1.5\%$, deflection falls below the allowable limit for all concrete strengths. The deflection reduction from increasing ρ is most pronounced at low reinforcement ratios, where the section transitions from heavily cracked to moderately cracked.

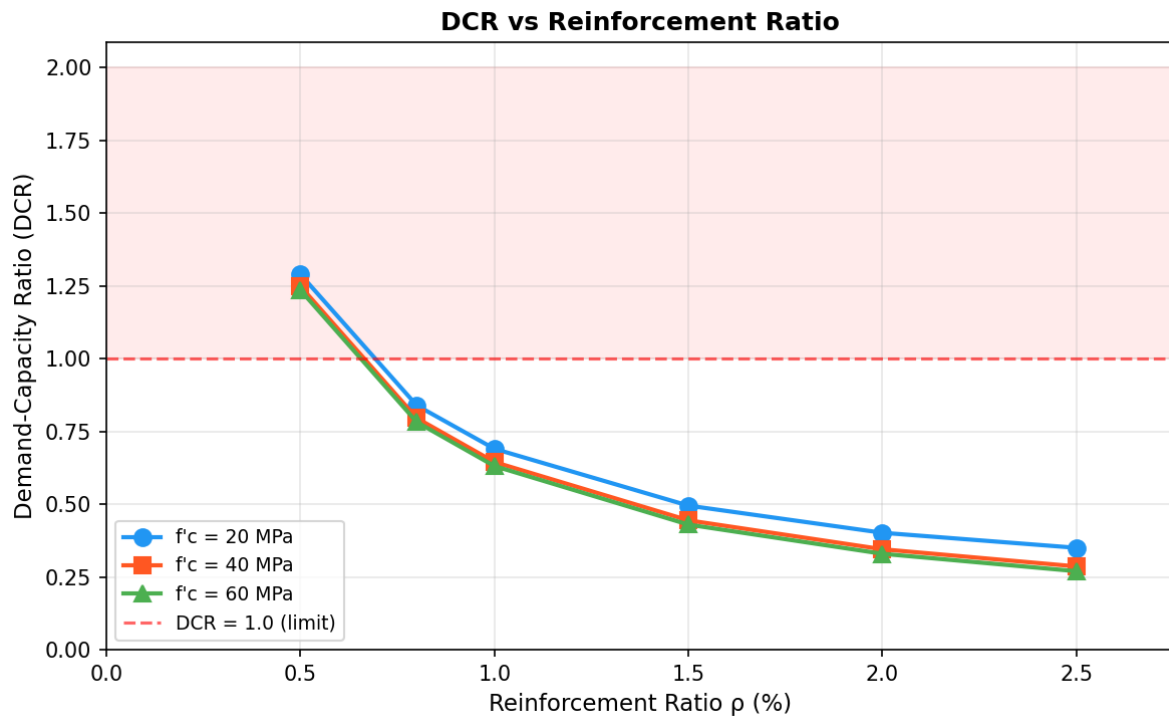


Figure 11: DCR vs Reinforcement Ratio

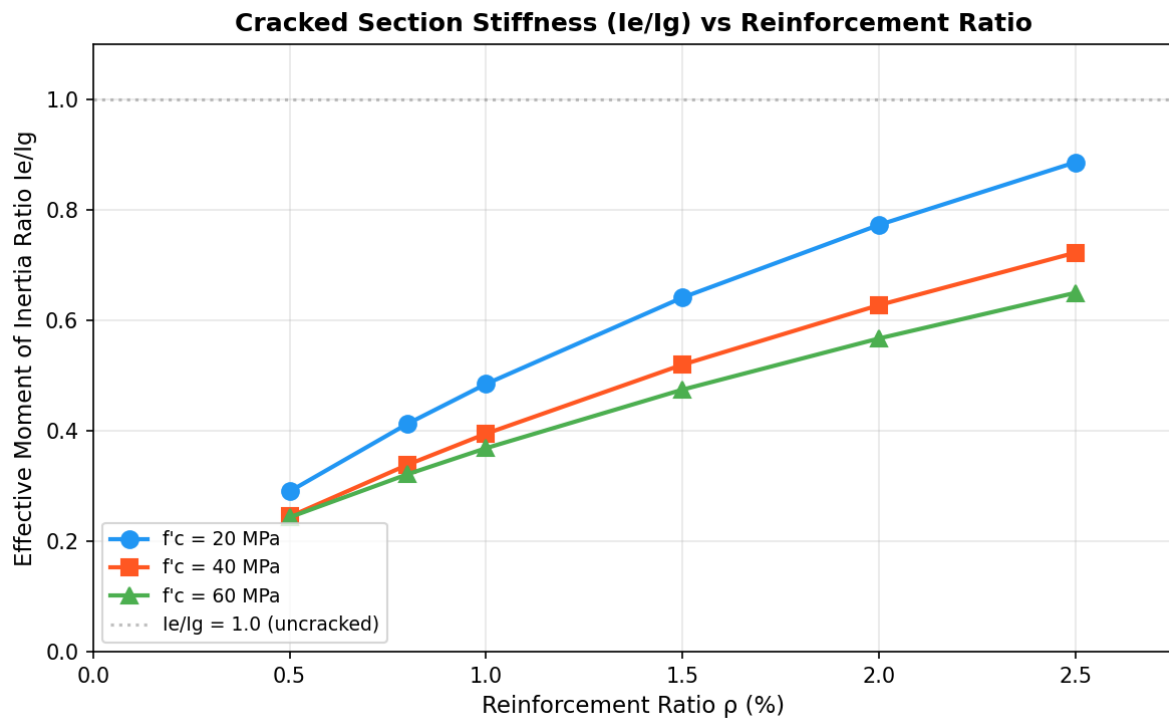


Figure 12: I_e/I_g vs Reinforcement Ratio

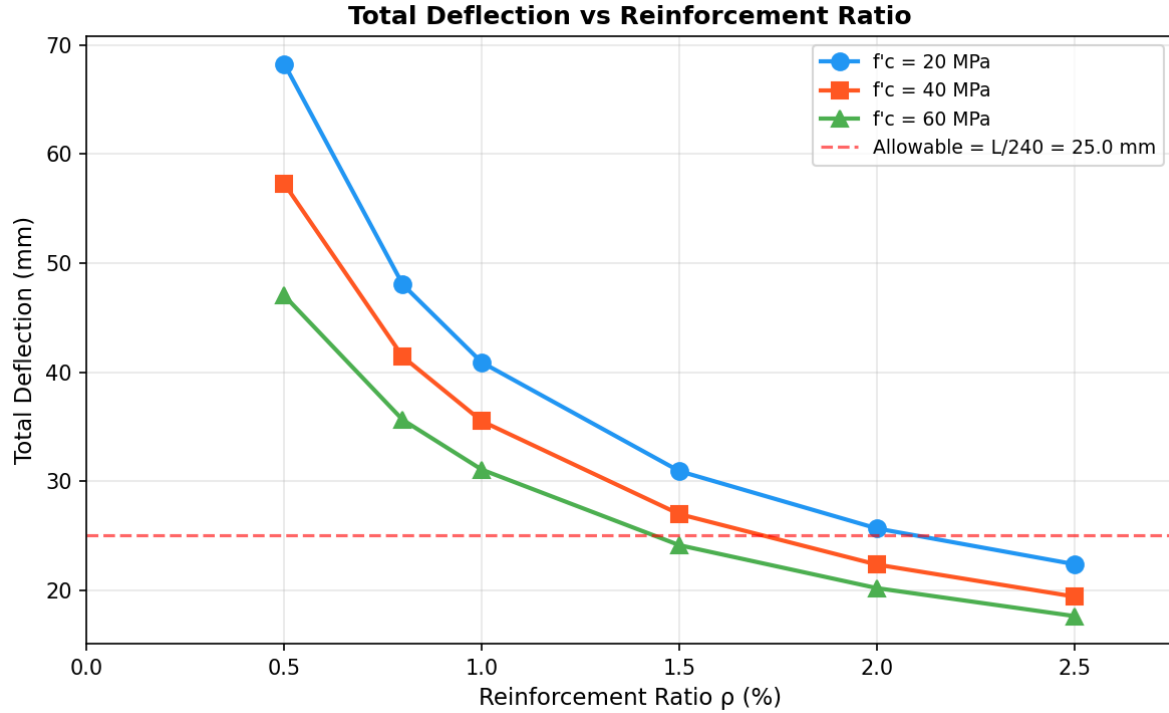


Figure 13: Deflection vs Reinforcement Ratio

1.8 7. Discussion

1.8.1 7.1 Validation Quality

The experimental validation demonstrates that the ACI 318-14 implementation in the Python framework produces predictions consistent with published code-based design expectations:

- **Flexure (nominal):** Mean $c = 1.109$, CoV = 5.85% — highly reliable
- **Shear with stirrups:** Mean $c = 1.089$, CoV = 21.53% — accurate but high scatter
- **Shear without stirrups:** Mean $c = 1.368$, CoV = 26.07% — very conservative

The high CoV for shear predictions is consistent with findings reported by Tošić et al. (2016) and reflects inherent limitations in the simplified ACI 318-14 shear equation, which does not account for aggregate interlock, dowel action, or size effects.

1.8.2 7.2 Implications for Design Practice

The parametric study results provide practical guidance for RC beam proportioning. For a beam subjected to typical gravity loads ($M_u = 200$ kN-m, $V_u = 100$ kN), a reinforcement ratio of 1.0% provides adequate strength with $DCR < 0.7$, but 1.5% may be needed to satisfy deflection limits. This highlights the importance of serviceability checks in RC beam design—strength alone is often not the governing criterion.

The I_e/I_g ratio results are particularly informative for practicing engineers. At $\rho = 0.5\%$, the effective stiffness is only 29% of the gross section, meaning that design assumptions based on uncracked section properties would significantly underestimate deflections.

1.8.3 7.3 Limitations

Several limitations of the present study should be acknowledged:

1. **Validation database:** The NAC database (50 beams) is relatively small, particularly for shear with stirrups (8 beams). Larger validation studies are needed.
 2. **Parameter range:** The parametric study covers only one beam geometry (300×600 mm) with constant loads. Results may differ for other geometries and loading conditions.
 3. **Deflection validation:** The deflection calculations have not been validated against experimental data, as the Tošić et al. (2016) database does not include deflection measurements.
-

1.9 8. Conclusions and Future Work

1.9.1 8.1 Conclusions

This paper has presented a comprehensive Python-based framework for RC beam design, validation, and parametric analysis. The key findings are:

1. **Design engine accuracy:** The ACI 318-14 implementation produces results consistent with hand calculations (error $< 0.1\%$) and validated against 50 experimental beams.
2. **Validation results:** ACI 318-14 nominal flexural predictions show mean $c = 1.109$ with $\text{CoV} = 5.85\%$. Shear predictions are more conservative but show higher variability ($\text{CoV} = 21\text{-}26\%$).
3. **Parametric insights:** Reinforcement ratio is the dominant factor controlling flexural capacity and deflection. The I_e/I_g ratio increases from 0.29 to 0.89 as ρ increases from 0.5% to 2.5%.
4. **Practical tool:** The framework provides a complete, transparent, and free alternative to commercial software for RC beam design, with an interactive web interface accessible to engineers without programming experience.

1.9.2 8.2 Future Work

Planned extensions to the framework include:

- **T-beam and L-beam sections:** Flanged section analysis per ACI 318-14 Section 6.3
 - **Seismic design (ACI Chapter 18):** Special moment frame provisions and ductility detailing
 - **Crack control (Chapter 24.3):** Flexural crack width verification
 - **RAC database expansion:** Incorporate recycled aggregate concrete beams from the same study for comparative analysis
 - **Optimization module:** Genetic algorithm for minimum-cost beam design
 - **PDF report generation:** Automated design reports with all calculations and plots
-

1.10 References

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1.11 Appendix A: Source Code Structure

All source code is organized under the project root:

```

scripts/
  RCBeam_moment_capacity.py      # Core design engine (design_beam function)
  app_beam_design.py             # Streamlit web application
  beam_diagram_calculator.py    # Shear/moment diagram calculator

research/scripts/
  validate_beam_aci.py           # NAC experimental validation
  rc_beam_comparison.py          # SPBeam comparison benchmark
  rc_beam_parametric_study.py    # Parametric study generator

research/data/nac_study/         # NAC experimental database (CSV files)
research/output/                 # Generated results (validation, comparison, parametric)
research/docs/                   # Research papers and documentation

```

1.12 Appendix B: Key Output Files

File	Description
research/output/nac_validation_report.pdf	Final validation report (18 flexure + 32 shear beams)
research/output/nac_validation_data	Python file for flexure validation data
research/output/nac_validation_data	Python file for shear validation data
research/output/beam_comparison_data.csv	SPBeam comparison data.csv

File	Description
research/output/parametric_study/phi_m_vs_rho.png	Study of ϕ_m vs ρ
research/output/parametric_study/Dphi/Dphi_vs_rho.png	Study of $D\phi$ vs ρ
research/output/parametric_study/Eigvals_rho.png	Study of Eigenvalues vs ρ
research/output/parametric_study/deflection_vs_rho.png	Study of deflection vs ρ